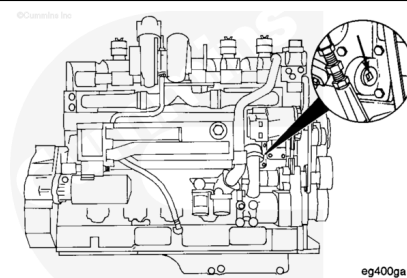


Trainings ▾

Preparatory Steps

⚠ CAUTION ⚠

The lubricating oil system must be primed before operating the engine after rebuild to avoid internal component damage. Do not prime the system from the bypass filter as the filter will be damaged.



All QSK45 and QSK60 industrial application engines are equipped with an automatic prelube device. If there is **not** an automatic prelube device on the engine, follow the prelube instructions.

For engines without an automatic prelube device, use a pump capable of supplying 205 kPa [30 psi] continuous pressure.

Remove the pipe plug at the front rear underside of the one-piece lubricating oil filter head.

Install an extended elbow (1) into the rear port the pipe plug was removed from. The final position of this elbow is facing the rear of the engine and slightly inboard, to make sure the check valve clears the front filter on the one-piece lubricating oil filter head.

Tighten the elbow.

Torque Value: 60 n·m [44 ft-lb]

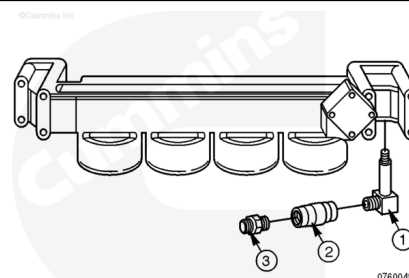
Apply a small amount of Loctite™ 641 to the tapered threads of the male adapter (3) and install it into the flow end of the check valve (2).

Apply a small amount of Loctite™ 641 to the tapered threads of the extended elbow (1) and install the check valve with the arrow pointing to the elbow in the lubricating oil filter head.

Tighten the check valve.

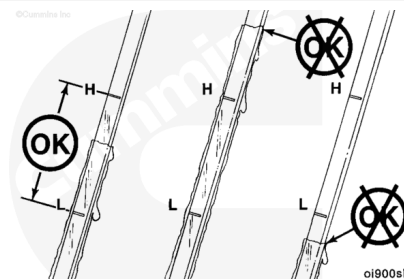
Torque Value: 48 n·m [35 ft-lb]

Connect the pump to the male adapter.



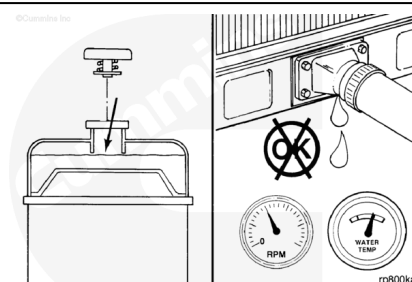
Use a supply of clean oil. Turn the pump to the ON position. Check the engine oil pressure gauge. When the gauge indicates oil pressure, begin monitoring the oil level in the oil pan.

Check the engine lubricating oil level to make sure it is filled to the proper level.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant or spray can cause personal injury.



⚠ CAUTION ⚠

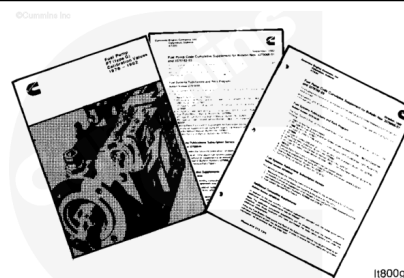
Do not add cold coolant to a hot engine. This can cause engine casting damage. Allow the engine to cool to below 50°C [120°F] before adding coolant.

Check the engine coolant level. Refer to Procedure 008-018 (Cooling System) in Section 8.

(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)

Use a known source of good quality number 2 diesel fuel. Number 1 diesel fuels, along with most other alternate fuels, are lighter (lower specific gravity, higher API gravity) than number 2 diesel fuel. The lighter the fuel, the lower the energy content per gallon (liter).

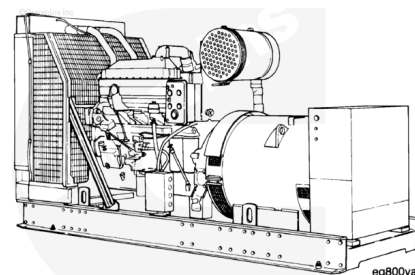
Engine operating specifications can be found in publications available from Cummins® Authorized Repair Locations.



Run-In Instructions

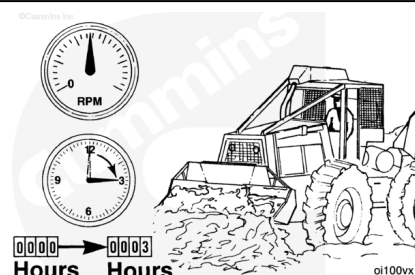
Generator Set Applications

Operate the engine in steps, varying the load from 25 percent to 100 percent, until blowby remains constant.



Industrial Applications

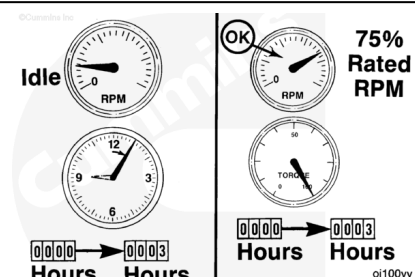
Operate the equipment in the normal duty cycle at partial load during the first three hours after rebuild.



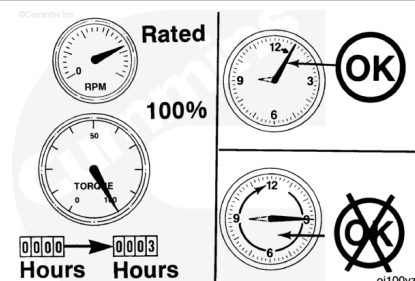
Do **not** idle the engine for more than 5 minutes at any one time.

Do **not** operate the engine at rated rpm. Operate at 75 percent of rated rpm or lower.

Rated rpm	Maximum rpm for First Three Hours
1900	1425
1800	1350



Do **not** operate the engine at full load for more than 5 minutes at any one time.



Finishing Steps

Remove the pump.

Remove the elbow (1), check valve (2), and adapter (3) from the lubricating oil filter head.

Install the pipe plug.

